HTE staff / Bench sheet

One semi-automated run · 48 pairs · 30 ligands

Staff weigh; the robot adds stock solvent. Staff move labware and handle caps, then take the final plate to HPLC-MS. 100 µL reactions: 0.1 M SM, 6 mol% each ligand, 12 mol% Pd, 10 mol% naphthalene. 95 °C / 500 rpm / nominal 4 h.

Slot	Load
1	HTEL stock holder: map on page 2. Moves 1 → 10 → 1 before dosing.
2	Axygen 12-channel, 12 mL/channel, high-profile V-bottom reservoir.
3	Para-Dox block with 48 empty reaction vials.
5, 8	300 µL tips for LEFT p300 single GEN2.
6	VWR 96 deep-well, 2.2 mL, V-bottom square-well LC plate.
9	300 µL tips for RIGHT p300 eight-channel GEN2.
10	Heater-shaker / universal flat adapter. Slots 4, 7 and 11 empty.

Bulk liquids / reservoir

Quench, 40 mL: 36 mL ACN + 3.95 mL water + 50 µL formic acid. **LC diluent, 60 mL:** 30 mL ACN + 29.94 mL water + 60 µL formic acid.

Channels	Liquid	Load per channel
A1-A2	Toluene	9 mL
A3-A5	Quench	9 mL
A6-A10	LC diluent	10 mL
A11 / A12	ACN / empty	3.5 mL / empty

Follow these steps / same numbers in the staff guide and App

Step	Staff action	Robot / result
1	Weigh into labelled vials (page 2). Make bulk liquids above. Load weighed chemicals with **no stock solvent added**.	Stock solvents are added by the robot.
2	Load deck. Open A1-F5 and A6/B6; keep ten analytical vials capped. Check offsets, 150 µL source pickup and both capped assemblies ≤45 mm. Import **run_screen.py**.	Follow staff-guide setup checks, then start. Numbered pauses give the next action.
3	At move prompts: **stock holder 1** → **10** → **1**. Inspect stocks; leave reaction sources open.	Makes/mixes stocks; shakes 750 rpm / 2 min, heater OFF; doses 48 reactions.
4	Cap all **48 reaction vials**. Cover reservoir/LC. Move **reaction block 3** → **10**. Start timer at hold-start prompt.	95 °C / 500 rpm; nominal 4 h heated hold.
5	After heating stops: **sealed block 10** → **3**. Uncover reservoir/LC; open ten analytical vials. Reaction caps stay on.	Makes seven standards A7-G7 while sealed reactions cool. LC plate stays in 6.
6	Operator decides when safe to open and module ready for unheated workup. Uncap reactions; **open block 3** → **10** → **3** at prompts.	400 µL quench/vial (500 µL total); 750 rpm / 2 min, heater OFF; settle 15 s.
7	No manual pipetting or labware move. Keep LC plate in slot 6.	Per column: eight fresh-tip 20 µL aliquots FIRST, then 480 µL diluent/well (8-channel, 2 × 240 µL).
8	Recap reaction vials for storage. Move **uncovered LC plate 6** → **10**, then remove when asked.	Final LC plate mix: 750 rpm / 2 min, heater OFF.
9	Seal/label plate and mark A1. Finish robot program. Carry flat to **HPLC-MS with lc-injections.csv**.	Medium-polarity 3-min LC: check 7 standards FIRST, then 48 samples. 165 min + overhead.

Timing: longer holds and pre-quench waits are allowed/logged. For a longer hold, remain at 4b until the agreed endpoint; heating/shaking stay on there. Return LC/MS files, actual injection map/method, yields/QC, robot log and run notes.

Weighing / stock-holder positions

All positions are in the HTEL stock holder in **slot 1**. Ligands occupy columns 1-5. Column 8 stays empty. The robot adds solvent directly in these vials. No stock transfers or refills.

Vial	Label	Material	mg
A1	L01	MorDal Phos	2.4
B1	L02	racBINAP	3.2
C1	L03	RockPhos	2.1
D1	L04	Alphos	4.2
E1	L05	AdBrettPhos	4.6
F1	L06	Me4t-BuXPhos	2.2
A2	L08	t-BuDave Phos	2.0
B2	L09	Xantphos	2.6
C2	L10	OTIPS DalPhos	2.1
D2	L11	Me3OMe-t-BuXPhos	2.0
E2	L12	VBRIDP	2.0
F2	L13	Trimesityl phosphine	2.0
A3	L14	CataCXium ABn	2.0
B3	L15	dppb	2.8
C3	L16	dcypf	3.8
D3	L17	t-BuBrettPhos	3.2
E3	L18	dtbpf	3.8
F3	L19	dppp	2.4
A4	L20	dppf	3.3
B4	L21	Xphos	2.0
C4	L22	RuPhos	2.1
D4	L23	SPhos	2.0
E4	L24	BrettPhos	2.4
F4	L25	MeDalPhos	2.5
A5	L26	Taniaphos SL-T001-2	3.5
B5	L27	tris(3-chlorophenyl)phosphine	2.7
C5	L28	BippyPhos	4.4
D5	L29	meCgPPh	2.0
E5	L30	N-XantPhos	2.5
F5	L31	Trixie Phos	2.0
A6	SM_IS	Pyridine-2-sulfonyl fluoride + naphthalene	96.7 SM +7.7 IS
B6	Pd	Pd(COD)(DQ)	27.3
C6	IS_MASTER	Naphthalene (analytical master)	10.0
D6	SM_FIT	Pyridine-2-sulfonyl fluoride	5.0
E6	SM_CHECK	Pyridine-2-sulfonyl fluoride	5.0
F6	P_FIT	2-Fluoropyridine	5.0
A7	P_CHECK	2-Fluoropyridine	5.0

Five empty working vials: B7 HIGH, C7 LOW, D7 CHECK, E7 IS_WORK, F7 MATRIX. **FIT** supplies the LC calibration; separately weighed **CHECK** sources supply an LC accuracy check. The robot makes all dilutions. Weigh liquid product by difference. Weigh ligands in the glovebox and retain original reagent bottles there.

Full procedure, maps, Python files and LC instructions: github.com/MiquelAngelPerezPuigdo/Agentic-AI-for-Dual-Ligand-Discovery / run